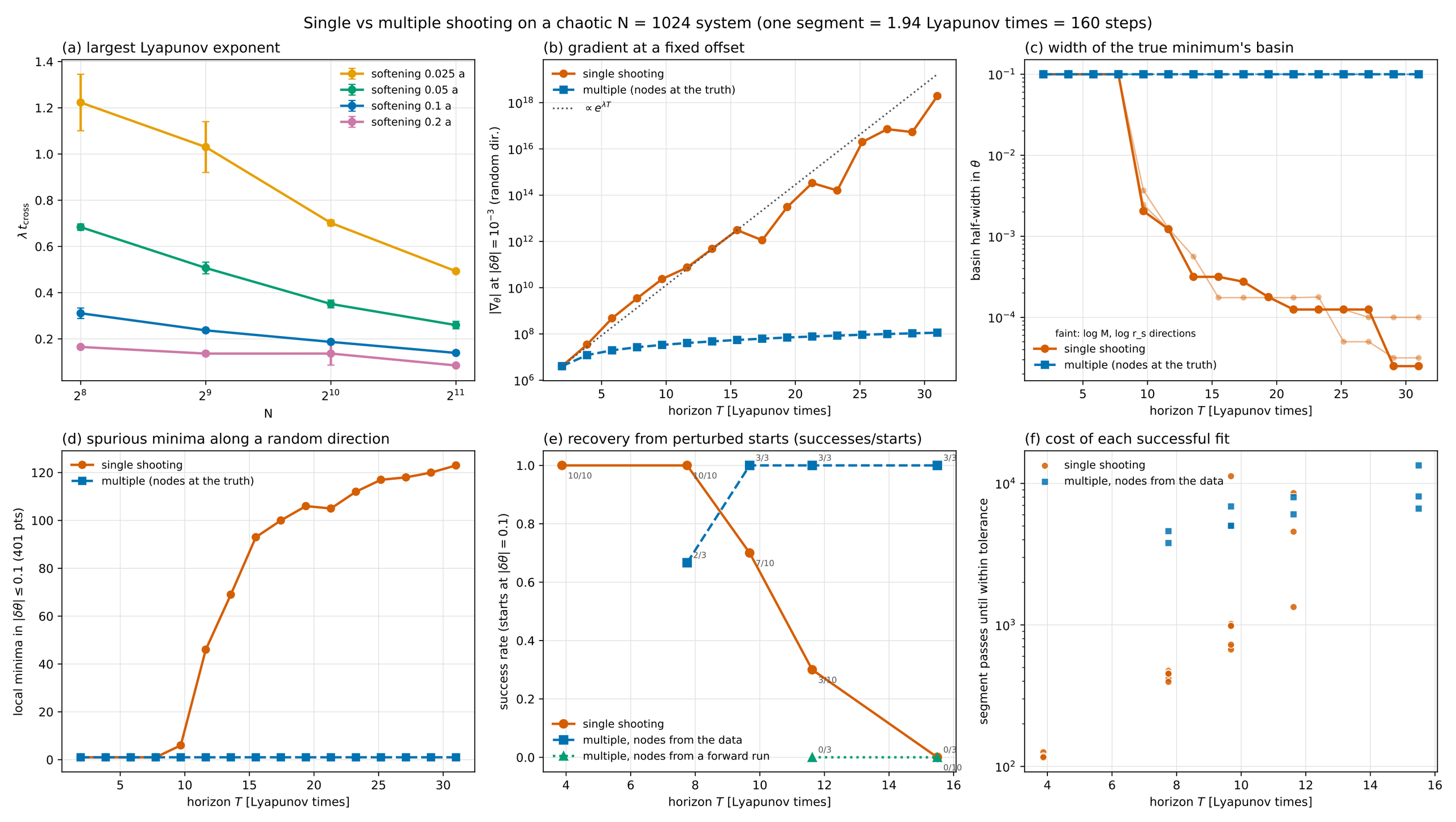
Single vs multiple shooting on a chaotic N = 1024 system (one segment = 1.94 Lyapunov times = 160 steps)
(a) largest Lyapunov exponent
softening 0.025 a
softening 0.05 a
softening 0.1 a
softening 0.2 a
$\lambda\, t_{cross}$
N
(b) gradient at a fixed offset
single shooting
multiple (nodes at the truth)
$\propto e^{\lambda T}$
$|\nabla_\theta|$ at $|\delta\theta| = 10^{-3}$ (random dir.)
horizon T [Lyapunov times]
(c) width of the true minimum's basin
basin half-width in θ
faint: log M, log r_s directions
single shooting
multiple (nodes at the truth)
horizon T [Lyapunov times]
(d) spurious minima along a random direction
single shooting
multiple (nodes at the truth)
local minima in $|\delta\theta| \le 0.1$ (401 pts)
horizon T [Lyapunov times]
(e) recovery from perturbed starts (successes/starts)
10/10
10/10
3/3
3/3
3/3
2/3
7/10
3/10
0/3
0/3
0/10
single shooting
multiple, nodes from the data
multiple, nodes from a forward run
success rate (starts at $|\delta\theta| = 0.1$)
horizon T [Lyapunov times]
(f) cost of each successful fit
single shooting
multiple, nodes from the data
segment passes until within tolerance
horizon T [Lyapunov times]